

SPIRE

A Contested-Logistics Decision Engine for the Stand-in Force

BLUF. A Marine S-4 forward in the First Island Chain today cannot see Class I, V, and VIII supply on one screen — those classes live in three separate systems while GCSS-MC handles Class IX parts. SPIRE collapses every Marine logistics feed into one role-shaped view, runs offline on a backpack-portable AI device (NVIDIA GB10, 128 GB unified memory, ~\$4,000, 2.6 lbs), records every decision in an audit trail that cannot be quietly edited, and shares with coalition partners in one click. **TRL 4, 1st place at Modern Day Marine 2026 (30 APR 2026). Operational pilot is the next step.**

The Operator's Problem

Today, a forward unit taking a Class VIII demand spike under degraded SATCOM has three systems open and a phone call. The planner has minutes — not hours — to choose between cross-level, expedite, or emergency requisition. The decision is unauditable; the artifact is a spreadsheet.

Capability Gap

“GCSS-MC now primarily focuses on requisitioning Class IX repair parts.” Classes I, V, and VIII are managed by “three separate incompatible systems.” “Conventional sustainment remain[s] the SIF concept’s greatest vulnerability.”

— *Sustainment of the Stand-in Force*, War on the Rocks (Sept 2022). The same fragmentation is acknowledged in the *Marine Corps Concept for Logistics* (Dec 2024).

What SPIRE does

Four role-shaped surfaces over one common picture of the unit:

- **Decision Bridge.** The home view. Five live signals on one screen: mission clock and FPCON, top alerts, forecasted Class I / V / VIII / IX shortages, mission-capable rate by unit with seven-day trends, and the health of the audit log itself.
- **PULSE.** Forecasts where shortages will hit and when, then drafts ranked options for the planner — cannibalize, expedite, or cross-level — sorted by which gets readiness back fastest. Predicts equipment failures days ahead and drafts the requisition before the asset breaks.
- **BASTION.** Common operating picture on a real geographic basemap with standard military symbols. Fuses access-control, drone, base-utility, and weather sensors into one threat picture. The operator drives the map by typing what they want to see.
- **SENTRY.** Automated classification of every record, with operator override at every step. Sanitized exports and one-click coalition release for FVEY, Japan, Australia, and the Philippines — the receiving partner verifies they got exactly what was sent.

Under the hood

- **Accepts data from any common source** — file drops, secure transfer, email, polled web services, polled databases, streaming feeds. The four core Marine logistics systems (GCSS-MC, DRRS-MC) are pre-mapped. An on-board AI model handles unfamiliar files; mappings save per unit so the work isn’t redone.
- **Audit by design.** Every change is cryptographically chained so an inspector can prove offline nothing has been altered. Feeds existing security monitoring tools.
- **Runs offline.** Fits on a backpack-portable AI device (NVIDIA GB10, 2.6 lbs, ~\$4,000). Decisions made forward during a SATCOM blackout reconcile automatically when comms restore. No cloud, ever.

Maturity & Risk

TRL 4. Validated at MDM 2026 against a synthetic GCSS-MC dataset built to mirror the live export shape. The next step — TRL 5 — requires a sanitized real GCSS-MC export from a sponsoring MEF G-4 or MLR S-4. No ATO yet; the system is built to IL5 standards (tamper-evident audit, role-based access, smartcard authentication, no cloud) but is not formally accredited.

Why Now

The 39th Commandant has named logistics the **pacing function**. The *Marine Corps Concept for Logistics* (Dec 2024) calls for resilience through redundancy, dispersion, and innovation. The data-fragmentation gap is named in published doctrine. The window to put SPIRE in operator hands is the FY27 POM cycle.

Path Forward — Ask

1. **90-day operational pilot** with one MEF G-4 or MLR S-4 willing to point a sanitized GCSS-MC export at the ingest path.
2. **An MCWL evaluator** to red-team SENTRY against operator-malpractice patterns.
3. **A POC at MCSC PM-MC2I or DC I&L** to translate operator requirements into a contract vehicle (SBIR Phase II or OTA).

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Repo: github.com/jeranaia/spire (public) · `docker-compose up` — clean laptop to live demo in under 5 minutes

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